



FACULTY OF ENGINEERING TECHNOLOGY

Subject: ~~computer~~ communication

Mid Exam Spring 24/25

Name: ~~XXXXXXXXXX~~

Duration: 50 min

EXAM CONTAINS TWO QUESTIONS

Q1: (Please write your answers in a table only, otherwise, your answers will be neglected)

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	C	B	D	D	A	C	B	A
Question	11	12	13	14						
Answer	A	D	A	C	H	A	T	R	O	N

4. Error Control and Flow Control are the responsibilities of
- data link and network Layers
 - Data Link and Transport Layer
 - presentation Layer
 - data link and Physical Layer
2. The SNR for ideal channel is:
- 0
 - 0.5
 - 1
 - ∞
3. Queuing time is function of:
- load
 - distance
 - transmission
 - phase
4. The bandwidth of digital signals used in data communications is:
- discrete with finite rate
 - discrete with an infinite range
 - continuous with finite range
 - continuous with an infinite range
5. We send a voice signal from a microphone to a recorder, the transmission is:
- Baseband transmission
 - Broadband transmission
 - only voice, no band
 - both a and b

6. The term that describes the position of the waveform relative to time 0 is
a) Wavelength b) Period c) Frequency ☒ d) Phase
7. The transmission impairment that refers to a signal with high energy in a very short time is:
a) Induced Noise b) Crosstalk c) Impulse Noise d) response noise
8. A signal with 200 milliwatts power passes through 10 devices each with an average noise of 2 microwatts. Calculate SNR:
a) 5000 b) 10000 c) 150 d) 250
9. Route determination can be identified by:
(a) Transport layer b) Application layer c) Network layer d) Physical Layer
10. A signal in which 1 bit lasts 0.001 s, the Bit rate would be:
a) 1kbps b) 500bps c) 1700bps d) 1Mbps
11. National Internet Service Provider (ISP) networks are connected to one another by private switching stations called:
☒ a) Network Access Points
b) Peering Points
c) National ISP
d) direct ISP
12. OSI stands for:
a) open system Internet
b) ~~Online Systems~~ Interconnection
c) Online systems Interconnection
d) ☒ Open Systems Interconnection
13. Keyboard and traditional monitors are examples of:
☒ a) Simplex devices
b) Duplex devices
c) Half Duplex devices
d) Full Duplex system
14. Two computers connected by an Ethernet hub are of:
a) TAN topology
b) MAN topology
c) WAN topology
d) ☒ LAN topology

Q2: A nonperiodic composite signal has a bandwidth of 240 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal

$$\text{Bandwidth} = \text{High} - \text{Low}$$

$$240 = 260 - 20$$

$$B = 240$$

$$L = 20 \quad 240 = 260 - 20$$

$$\text{High} = 260 \text{ kHz}$$

$$\text{Low} = 20 \text{ kHz}$$

